| Qn | | | Answers | Marks |
|---|---|---|---|---|
| 8 | (a) | (i) | Electric field is directed downwards; for the oil drop to remain stationary / Electric field points in direction of decreasing potential | M1 |
| | | | the electric force must act upwards, hence charge is negative | A1 |
| | | (ii) | $E = \frac{V}{d} = \frac{680}{7.0 \times 10^{-3}} = 9.71 \times 10^4 \text{ V m}^{-1}$ | C1 |
| | | | $qE = 4.8 \times 10^{-14}$<br>$q = \frac{4.8 \times 10^{-14}}{9.71 \times 10^4}$ | M1 |
| | | | $q = -4.94 \times 10^{-19}$ C<br><br>e.c.f from (a)(i) for sign of charge | A1 |
| | | (iii) | $m = \frac{4.8 \times 10^{-14}}{9.81} = 4.89 \times 10^{-15}$ kg<br><br>$F_E - W = ma$<br>$\frac{qV}{d} = W + ma$ | M1 |
| | | | $V = \frac{(W + ma)d}{q}$<br><br>$V = \frac{(4.8 \times 10^{-14} + 4.89 \times 10^{-15} \times 0.50)(7.0 \times 10^{-3})}{4.94 \times 10^{-19}}$ | M1 |
| | | | $= 710\ V \quad (714.8)$ | A1 |

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|---|---|---|---|---|
| | **(b)** | **(i)** | If *B* is into the page, then *E* is downwards.<br><br>OR<br><br>If *B* is out of the page, then *E* is upwards | A1 |
| | | **(ii)** | $E = vB = (1.6 \times 10^5)(9.7 \times 10^{-2})$ | M1 |
| | | | $E = 1.55 \times 10^4 \text{ V m}^{-1}$ | A1 |
| | | **(iii)** | Particles will experience a <u>larger electric force</u>, only particles that <u>move faster will have a larger magnetic force</u> such that net force is zero. | M1 |
| | | | Therefore, particles with <u>a greater speed pass undeviated</u>. | A1 |
| | **(c)** | **(i)** | A <u>current-carrying wire in a magnetic field</u> experiences <u>a magnetic force. / Interaction between the magnetic field produced by the current and that of the magnet results in a force between the wire and magnet.</u> | B1 |
| | | | The magnet experiences a <u>upward force by the wire, reducing the force exerted on the balance</u> and hence the balance reading. | B1 |
| | | **(ii)** | **1.**<br>For the magnet in equilibrium,<br>$W = F_B + N$<br>$W = BL(2.0) + 2.20$<br>$W = BL(3.5) + 2.14$ | C1 |
| | | | $BL = 0.04\ N\ A^{-1}; W = 2.28\ N$ | C1 |
| | | | $M = 0.232 \text{ kg}$ | A1 |
| | | | **2.** The <u>magnetic force on wire is downwards</u> according to Newton's 3rd law. By <u>Fleming's left hand rule</u>, the direction of the current | M1 |
| | | | must be from B to A. | A1 |